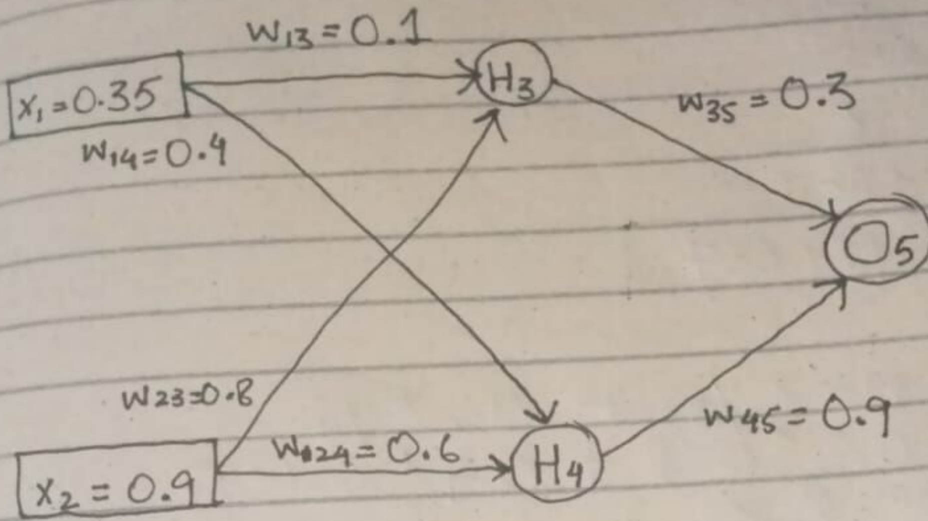
Given a Neural Network:

x = inputs , w = weights , y = output



Actual Output is 0.5

Learning Rate is 1

Required:

1. Perform a forward and backword pass
2. Perform another forward pass

FORWARD PASS 1:

$$x_3 = w_{13} * x_1 + w_{23} * x_2$$
$$= 0.1 * 0.35 + 0.8 * 0.9$$
$$= 0.755$$

$$y_3 = \frac{1}{1+e^{-x}}$$
$$= \frac{1}{1+e^{-0.755}}$$

$$\boxed{y_3 = 0.68}$$

$$X_4 = W_{14} * x_1 + W_{24} * x_2$$
$$= 0.4 * 0.35 + 0.6 * 0.9$$
$$= 0.68$$

$$Y_4 = \frac{1}{1+e^{-x}}$$

$$= \frac{1}{1+e^{-0.68}}$$

$$\boxed{Y_4 = 0.6637}$$

$$X_5 = W_{35} * Y_3 + W_{45} * Y_4$$
$$= 0.3 * 0.68 + 0.9 * 0.6637$$
$$= 0.8013$$

$$Y_5 = \frac{1}{1+e^{-x}}$$

$$= \frac{1}{1+e^{-0.8013}}$$

$$\boxed{Y_5 = 0.69}$$

BACKWARD PASS :

$$Error = Y_{actual} - Y_5$$
$$= 0.5 - 0.69$$
$$= -0.19$$

$$\delta_5 = 0.69 (1-0.69) (-0.19)$$
$$\boxed{\delta_5 = -0.04.}$$

$\delta_3 = 0.68\,(1-0.68)\,[0.3 * (-0.04)]$

$\boxed{\delta_3 = -0.003}$

$\delta_4 = 0.6637\,(1-0.6637)\,[0.9 * (-0.04)]$

$\boxed{\delta_4 = -0.008}$

Δw_{ij}	Calculate	Δw_{ij}	w_{ij} update
Δw_{13}	$y_1 \delta_3$	-0.0011	0.0989
Δw_{14}	$y_1 \delta_4$	-0.0028	0.3972
Δw_{23}	$y_2 \delta_3$	-0.0027	0.7973
Δw_{24}	$y_2 \delta_4$	-0.0072	0.5928
Δw_{35}	$y_3 \delta_5$	-0.0272	0.2728
Δw_{45}	$y_4 \delta_5$	-0.0265	0.8735

FORWARD PASS 2 :

$x_3 = w_{13} * x_1 + w_{23} * x_2$

$= 0.0989 * 0.35 + 0.7973 * 0.9$

$x_3 = 0.7522$

$y_3 = \dfrac{1}{1+e^{-x}}$

$= \dfrac{1}{1+e^{-0.7522}}$

$\boxed{y_3 = 0.6797}$

$x_4 = w_{14} * x_1 + w_{24} * x_2$
$= 0.3972 * 0.35 + 0.5928 * 0.9$
$= 0.6725$

$y_4 = \frac{1}{1+e^{-x}}$
$= \frac{1}{1+e^{-0.6725}}$
$\boxed{y_4 = 0.6621}$

$x_5 = w_{35} * y_3 + w_{45} * y_4$
$= 0.2728 * 0.6797 + 0.8735 * 0.6621$
$= 0.7638$

$y_5 = \frac{1}{1+e^{-x}}$
$= \frac{1}{1+e^{-0.7638}}$
$\boxed{y_5 = 0.6821}$